AMARNATH S. PATEL

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EDUCATION

University of Central Florida — **Undergraduate Student** **4.00 GPA**

B.S. in Physics (Optics & Lasers) and Mathematics, Computer Science Minor — August 2025 – Present

Florida Atlantic University — **3.66 GPA**

Computer Science coursework concurrent with High School Diploma (111 Credit Hours) — August 2021 – May 2025

RESEARCH EXPERIENCE

UCF Astrophotonics Lab, CREOL — Undergraduate Researcher — August 2025 – Present

Supervisor: Dr. Stephen Eikenberry

- Developing instrumentation and automation software for astrophotonics research at the intersection of photonics and observational astronomy, spanning CREOL and the UCF Physics Department.
- Built an automated data acquisition and processing pipeline for modal characterization of photonic lanterns via off-axis digital holography; coordinates 4 instruments across 7-leg × 7-wavelength (1540–1570 nm) sweeps, implementing FFT-based twin-image extraction, Butterworth filtering, quadratic phase correction, and LP mode decomposition to recover complex modal amplitudes and field structure.
- Developing PolyOculus, control and automation software for an 8-telescope photometric observation array; implementing INDI-protocol mount control, focuser automation, RA/Dec coordinate slewing, and backlash compensation.
- Developed Python SDK wrapper (gentec-camera) for an IR beam profiling camera enabling automated acquisition, real-time beam analysis, and FITS output.

UCF Physics Department — Paid Undergraduate Research Assistant — March 2026 – Present

Physics Education Research — Supervisor: Dr. Zhongzhou Chen

- Developing ESTELA, an automated multi-version exam generation system for AI-assisted isomorphic physics problem banks, supporting scalable and equitable assessment infrastructure for introductory STEM courses.
- Built in Rust with a vanilla JS frontend; parses YAML problem banks across 13 topic areas supporting 11 question types, renders LaTeX math via KaTeX, and exports multi-version exams as LaTeX or print-ready HTML/PDF.
- Funded by NSF award 2421299 and Gates Foundation INV-076932.

FAU Grant-Funded AI Safety Research Project (High School) — January 2024 – March 2025

Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10+ coherence and detection methods (BERT NSP, GPT-2 perplexity, RoBERTa, LSA, NLI, burstiness) across 3,125+ generated sequences to evaluate approaches for identifying AI-generated text.
- Identified GPT-2 perplexity as the strongest discriminator (3.3× gap between coherent and incoherent text); BERT NSP failed to distinguish the two.
- Characterized a fundamental quality–evasion tradeoff across all humanization approaches tested, including iterative sampling, content abstraction, and RL-based methods.
- Grant-funded with HPC access; findings presented at the Wilkes Honors College Symposium (2025).

OTHER EXPERIENCE

Teaching Assistant — Calculus (High School) — August 2024 – May 2025

- Assisted 70 undergraduate students with learning calculus; office hours, exam review, grading. Part-time (10 h/week).

Advanced Experimental Vehicles — Programmer, Leader, Builder (High School) — November 2023 – May 2025

- Configured Arch Linux ARM on Raspberry Pi 5 with Hyprland compositor and WireGuard VPN, enabling worldwide real-time telemetry monitoring of BMS, GPS, and camera feeds.
- Won 2nd Place in Division and Lockheed Martin Award for "Highest Level of Engineering Excellence."

PRESENTATIONS

Coherence and Detection Approaches for Identifying AI-Generated Text 2025

Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

SELECTED PROJECTS

Photonic Lantern Digital Holography Automation January 2026 – Present

- Automated data acquisition and processing pipeline for photonic lantern characterization via digital holography. Orchestrates tunable IR laser, fiber switch, motorized polarization controller, and GigE InGaAs camera across 7-leg × 7-wavelength sweeps (1540–1570 nm). Implements FFT-based fringe extraction and full 7-mode LP decomposition.

PolyOculus **— 8-Telescope Array Control Software** January 2026 – Present

- INDI-protocol mount control, automated focuser, RA/Dec slewing, and backlash compensation for an 8-telescope photometric observation array. Part of ongoing astrophotonics research at UCF CREOL.

ESTELA — Problem Bank Visualizer & Exam Generator March 2026 – Present

- Rust backend, vanilla JS frontend. Parses YAML problem banks across 13 topic areas and 11 question types. Renders LaTeX via KaTeX. Exports multi-version exams as LaTeX or print-ready HTML/PDF. NSF/Gates-funded research tool.

gentec-camera **— IR Beam Profiling SDK** November 2025

- Python SDK wrapper for Gentec IR beam profiling camera. Automated acquisition, real-time Gaussian beam analysis, FITS output for optical instrumentation research.

TECHNICAL SKILLS

Programming Languages: C/C++, Rust, Python, JavaScript, Shell (Fish, Bash, tcsh), LaTeX/Typst

Tools & Libraries: NumPy, SciPy, Docker, Git, CMake, XMake, INDI

Instrumentation: Tunable IR laser control, GigE camera acquisition, fiber switch automation, motorized polarization control, FITS data format

Operating Systems: Linux (Arch, NixOS, Ubuntu), BSD, Windows

RELEVANT COURSEWORK

University of Central Florida

Physics

Geometric Optics & Lab (A)
Chemistry Fundamentals I (A)
Quantum Information Processing (A)
Directed Independent Research (IP)
General Physics III (A)
Chemistry Fundamentals II (IP)
Theoretical Methods for Physics (A)
Electricity & Magnetism I (IP)

Mathematics

Statistical Methods I (A)
Linear Algebra (IP)
Intro to Partial Differential Equations (IP)
Matrix & Linear Algebra (A)
Introduction to Complex Variables (IP)
Introduction to Graph Theory (IP)

Computer Science

Engineering Analysis & Computation (A)
Object-Oriented Programming (IP)
Discrete Structures (IP)

Florida Atlantic University

Physics

General Physics I (C+)
General Physics II — Honors (A)

Mathematics

Calculus I (A−)
Calculus II (A−)
Honors Calculus III (A)
Differential Equations (A)
Matrix Theory (B)

Computer Science

Data Structures & Algorithms (A)
Computer Logic Design (A)
Computer Architecture (A−)
Intro to Deep Learning (A)
Foundations of Computing (A)

HONORS & AWARDS

Lockheed Martin Award — "Highest Level of Engineering Excellence," AEV Competition 2024

2nd Place in Division, Advanced Experimental Vehicles Competition 2024

NSF Award 2421299 — Research Funding (ESTELA project, with Dr. Zhongzhou Chen) 2026

Gates Foundation INV-076932 — Research Funding (ESTELA project) 2026